

The Sociable Sciences: Darwin and His Contemporaries in Chile
By Patience A. Schell

The First Atomic Age: Scientists, Radiations, and the American Public, 1895-1945
By Matthew Lavine

NASA in the World: Fifty Years of International Collaboration in Space
By John Krige, Angelina Long Callahan, and Ashok Maharaj

Empire and Science in the Making: Dutch Colonial Scholarship in Comparative Global Perspective
Edited by Peter Boomgaard

Anglo-American Connections in Japanese Chemistry: The Lab as Contact Zone
By Yoshiyuki Kikuchi

Eismitte in the Scientific Imagination: Knowledge and Politics at the Center of Greenland
By Janet Martin-Nielsen

Climate, Science, and Colonization: Histories from Australia and New Zealand
Edited by James Beattie, Matthew Henry and Emily O'Gorman

The Surveillance Imperative: Geosciences during the Cold War and Beyond
By Simone Turchetti and Peder Roberts

Post-Industrial Landscape Scars
By Anna Storm

Voices of the Soviet Space Program: Cosmonauts, Soldiers, and Engineers Who Took the USSR into Space
By Slava Gerovitch

After Apollo?: Richard Nixon and the American Space Program
By John M. Logsdon

After Apollo?

Richard Nixon and the American Space Program

John M. Logsdon

palgrave
macmillan



AFTER APOLLO?

Copyright © John M. Logsdon, 2015.

All rights reserved.

First published in 2015 by

PALGRAVE MACMILLAN®

in the United States—a division of St. Martin's Press LLC,

175 Fifth Avenue, New York, NY 10010.

Where this book is distributed in the UK, Europe and the rest of the world, this is by Palgrave Macmillan, a division of Macmillan Publishers Limited, registered in England, company number 785998, of Houndmills, Basingstoke, Hampshire RG21 6XS.

Palgrave Macmillan is the global academic imprint of the above companies and has companies and representatives throughout the world.

Palgrave® and Macmillan® are registered trademarks in the United States, the United Kingdom, Europe and other countries.

ISBN: 978-1-137-43852-2

Library of Congress Cataloging-in-Publication Data

Logsdon, John M., 1937–

After Apollo? : Richard Nixon and the American space program / John M.

Logsdon.

pages cm—(Palgrave studies in the history of science and technology)

Includes bibliographical references and index.

ISBN 978-1-137-43852-2 (hardback : alk. paper)

1. Astronautics and state—United States—History—20th century.

2. United States. National Aeronautics and Space Administration—Planning—History—20th century. 3. Project Apollo (U.S.)—History. 4. Nixon, Richard M. (Richard Milhous), 1913–1994—Influence. I. Title.

TL789.8.U5L6385 2014

629.4500973'09047—dc23

2014035724

A catalogue record of the book is available from the British Library.

Design by Newgen Knowledge Works (P) Ltd., Chennai, India.

First edition: March 2015

10 9 8 7 6 5 4 3 2 1

Printed in the United States of America.

This book is dedicated to my grandchildren—Jacob, Sara, and Aaron—with the hope that they will see the space dream become reality. I was present when men first left for the Moon in 1969. I hope that they will see men and women return to the Moon and leave for Mars—or maybe even make the trip themselves.

Epilogue

Richard Nixon and the American Space Program

President Richard Nixon and his associates between 1969 and 1972 made three major decisions with lasting consequences for the U.S. space program.¹ The preceding chapters have chronicled the making of those decisions. This summary chapter will assess their character and discuss their impact on the U.S. space program over the more than four decades since they were made.

The three principal Nixon administration space policy decisions were:

- *To treat the space program, not as a special, high-priority government activity as had been the case during Apollo, but rather as part of the "day in and day out" activities of government, with its budget determined "within a rigorous system of national priorities."** The Nixon administration formalized NASA's need to compete with other government agencies through the political and budgetary processes for priority, and then assigned a relatively low priority to space activities in that competition.
- *To lower U.S. ambitions in space by not setting another challenging space goal and thus ending for the foreseeable future human space flights beyond low Earth orbit.* As Assistant to the President Peter Flanigan remarked at the time, there was in the White House in 1969 and early 1970 "a feeling that the country had had enough excitement [in space] for now"; there was no inclination on the part of Richard Nixon to propose another Kennedy-like space goal for the post-Apollo period or even to indicate in any but the most general terms that the United States would continue to work toward human exploration of the solar system.
- *To build the post-Apollo program around the space shuttle without linking the shuttle to a long-term strategy for its use.* The shuttle was seen as a new capability for carrying out the space program of the 1980s and beyond. Those directly involved in shuttle planning saw it as a first step

* Citations to material quoted in previous chapters will not be repeated here.

toward a comprehensive space exploitation and exploration capability. However, NASA did not clearly present this perspective to Richard Nixon and his associates as the space agency sought shuttle approval, and the Nixon administration did not couple its approval to a strategic perspective on long term space program goals. As historian Walter McDougall later observed, "Apollo was a matter of going to the moon and building whatever technology could get us there; the Space Shuttle was a matter of building a technology and going wherever it could take us."²

The first two of these decisions were made early in the Nixon administration, in the context of the White House quickly rejecting the ambitious post-Apollo space program proposed in the 1969 Space Task Group report. While these decisions were resisted by NASA, there was little controversy among Richard Nixon and his advisers in making them; their collective intent from the start of their time in the White House was to follow Apollo with a much more modest space effort. In contrast, the decision to develop the space shuttle was the end product of a contentious three-year decision process, with NASA pushing for approval of a technologically ambitious shuttle design and White House budget and technical advisers opposing such an undertaking and proposing more modest approaches to keeping human space flight a part of post-Apollo activities. Richard Nixon and his senior advisers gave little weight to economic and technical arguments, seeing the shuttle program primarily in a political context. The NASA position prevailed, with four-decade consequences for the U.S. space program.

The Space Program and National Priorities

Richard Nixon made it clear to his associates that he did not want the post-Apollo space effort appear to take money away from government programs on Earth. As the March 1970 statement outlining his space policy was being prepared, Nixon stressed that it should avoid "positive statements on space" being "invidiously" compared to his attitude toward "problems in poverty and social problems here on earth." He did not want to be seen as "taking money away from social programs and the needs of the people here [on Earth] to fund spectacular crash programs out in space." Nixon was a careful reader of opinion polls and other indications of public sentiment, and his generalized sense that space achievements were part of "exploring the unknown" did not override his sense that an ambitious space program was not something that would gain political support. He was not interested in leading the nation to accept an ambitious post-Apollo space effort.

This perspective was applied in an ad hoc fashion to budget decisions on the NASA program in December 1969 and January 1970, and space did not fare well as it competed for funding with other Nixon administration priorities in the overall context of an imperative to balance the federal budget. In parallel with the chaotic Fiscal Year 1971 budget process, there was a move to formalize the approach the Nixon administration would take

to setting the priority of post-Apollo space efforts. The result was what has been characterized in this study as the "Nixon space doctrine," clearly stated in the March 7, 1970, presidential statement on space. Characterizing this statement as a "doctrine" may be rather overstating the reality; it was more an after-the-fact rationalization of the perspectives that led Nixon and his associates to reject the recommendations of the Space Task Group and even in the aftermath of the Apollo success to continue to reduce the NASA budget. But in the sense that the framework for space decision making set out in the Nixon statement has in its essence been accepted by most presidents since, it can be said to deserve being called a "doctrine."

The Nixon space doctrine had two elements. The first was to change the status of the space program from an effort formally assigned the highest national priority, as had been the case during Apollo, to just one of many "normal" government activities. In the language of the space statement: "We must think of them[space activities] as part of a continuing process—one which will go on day in and day out, year in and year out—and not as a series of separate leaps, each requiring a massive concentration of energy and will and accomplished on a crash timetable." Space was to become "a normal and regular part of our national life."

The second element of the doctrine was to declare that the space program from 1970 forward would have to compete with other discretionary government activities for priority and corresponding budgetary support. The space statement said: "Space expenditures must take their proper place within a rigorous system of national priorities. What we do in space from here on in... must therefore be planned in conjunction with all of the other undertakings which are also important to us."

The Impact of the Nixon Space Doctrine

The proposition that the space program should not be based on "a series of separate leaps, each requiring a massive concentration of energy and will and accomplished on a crash timetable," has had a continuing impact on presidential decisions on the space program. President Jimmy Carter in 1978 approved a space policy statement that explicitly echoed the Nixon declaration; it said "our space policy will become more evolutionary rather than centering on a single, massive engineering feat."³ Even though most presidents since Richard Nixon have proposed some type of major new space development and in most cases provided a timetable for its achievement, in none of those proposals was the undertaking to be carried out on a "crash" basis, and certainly none were accompanied by a "massive concentration of energy and will," not to mention adequate financial resources.

The Nixon decision that "space expenditures must take their proper place within a rigorous system of national priorities" has had an even more lasting impact on the U.S. space program. At the peak of the Apollo buildup in 1966, the NASA budget comprised nearly 4.4 percent of federal spending overall and 19 percent of discretionary nondefense federal spending. (The

NASA share of the federal budget is most frequently cited in terms of a percentage of the overall budget. Given the inexorable growth of the portion of the budget devoted to mandatory entitlements, it seems more useful to discuss the NASA budget in terms of its share of the discretionary nondefense budget, since it is in that realm that space spending competes with other discretionary government programs.) As President Lyndon B. Johnson refused to approve any of NASA's post-Apollo proposals, NASA's budget share quickly began to decline from its Apollo high point; by the time Richard Nixon became president the NASA budget had dropped to just above 8 percent of discretionary nondefense spending. The early Nixon space decisions continued this trend; in Fiscal Year 1973, the budget in which space shuttle approval was first reflected, the NASA discretionary budget share was approximately 6 percent and on a downward trajectory. While it was under Lyndon Johnson rather than any of his successors that the biggest percentage reduction in NASA's budget share occurred, that reduction came from deferring a decision on what to do in space after Apollo, not on the basis of a specific decision to lower the space program's priority. By contrast, Richard Nixon consciously made that crucial decision—to reduce NASA's priority rather than assign it new, expensive programs and thus continuing rather than reversing the decline in NASA's budget share. The NASA portion of discretionary nondefense spending vacillated between 6 and 4 percent between 1977 and 2002 and between 4 and 3 percent since. By any measure, the space program has not done well in competition for budget resources; in fact, compared to other government programs, it has declined in priority over the past 40 years.⁴

The consequences of this declining share of the overall discretionary budget have been clear to most observers. For example, the Columbia Accident Investigation Board in 2003 observed that "NASA has had to participate in the give and take of the normal political process in order to obtain the resources needed to carry out its programs." In this give and take, "NASA has usually failed to receive budget support consistent with its ambitions. The result . . . is an organization straining to do too much with too little."⁵

Increasing the NASA Budget Share?

The reaction to this situation on the part of the space flight community has been predictable—continuing advocacy that the NASA budget share should be increased. A 1990 space program review led by aerospace industry executive Norm Augustine suggested that "a reinvigorated space program will require real growth in the NASA budget of approximately ten percent per year (through the year 2000), reaching a peak spending level of about \$30 billion per year (in constant 1990 dollars) by about the year 2000."⁶ A NASA FY2000 budget of \$30 billion in 1990 dollars would have been the equivalent of a budget of almost \$40 billion in 2000 dollars; the actual NASA budget for Fiscal Year 2000 was \$13.6 billion.⁷ Almost two decades later, a similar review of NASA's human space flight program,

again led by Augustine, reached only a slightly less ambitious conclusion, observing that "NASA's budget should match its mission and goals," but then suggesting that "meaningful human exploration" would be possible only if the NASA budget were increased by up to \$3 billion per year for several years.⁸ Given that the proposed NASA budget at the time the review was taking place was \$18.7 billion, this was a call for an over 15 percent increase in NASA's annual resources. More recently, astrophysicist and science spokesperson Neil DeGrasse Tyson has gained widespread attention by his advocacy of doubling the NASA budget, bringing it back to 1 percent of overall Federal spending. Such an action, suggests Tyson, would "give NASA enough money to do everything everyone has wanted NASA to do over all these years *and* enable us to go back to the moon and on to Mars in a bold and audacious way."⁹

All of these recommendations and suggestions fly in the face of a reality set in motion by the Nixon space doctrine: When the priority of the space program is compared through the normal political process to the priority of other uses of government funds, the outcome is to allocate to the space program a relatively low share of federal discretionary spending, inadequate to support a vigorous and sustainable program of space exploration. This outcome has been consistent for over 40 years and is very unlikely to change anytime soon. A 2014 review of the U.S. human space flight program observed that "human spaceflight—among the longest of long-term endeavors—cannot be successful if held hostage to traditional short-term decision-making and budgetary processes." But the Nixon space doctrine declared that it was through those processes that space budget decisions should be made. The same report also noted that "it serves no purpose for advocates of human exploration to dismiss these realities [the lack of public interest in space and the attendant low priority given to increasing space spending] in an era in which the citizenry and national leaders are focused intensely on the unsustainability of the national debt, [and] the growth in entitlement spending . . . There is at least as great a chance that human spaceflight budgets will be below the recent flat line trend as above it."¹⁰ The mismatch between the requirements of a successful program of human space exploration and the tenets of the Nixon space doctrine has been a central space policy reality since the doctrine was first stated in 1970.

Ending Exploration

Richard Nixon saw in the *Apollo 11* mission a unique opportunity. Project Apollo had been intended from its 1961 approval by President Kennedy to be a large-scale effort in "soft power," sending a peaceful but unmistakable signal to the world that the United States, not its Cold War rival the Soviet Union, possessed preeminent technological and organizational power.¹¹ Nixon agreed with this rationale for the lunar landing program, and in his first months as president made sure to identify himself and his foreign policy agenda with what he later would hyperbolically characterize as "the greatest

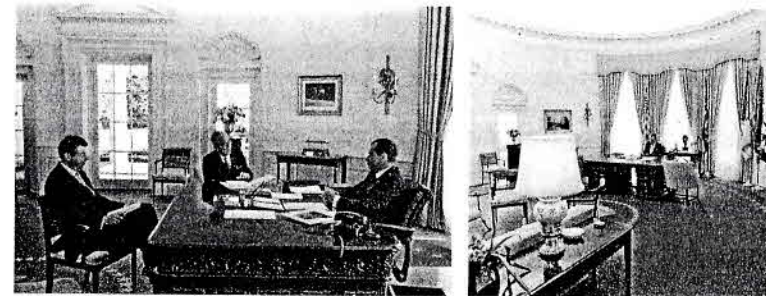
week in the history of the world since the Creation." But Nixon's embrace of Project Apollo as a tool of American soft power was short-lived. Once the United States had won the race to the Moon, he perceived little foreign policy benefit from subsequent lunar landing missions or from approving a post-Apollo program focused on preparing for missions to Mars. Other considerations, primarily domestic in character, would determine the Nixon approach to space in the post-Apollo period.

Like many other Americans, Nixon quickly lost interest in continuing Apollo flights to the Moon. As early as December 1969, after the first two lunar landings, he remarked that he "did not see the need to go to the moon six more times." When the *Apollo 12* crew visited the White House that month, mission commander Pete Conrad came away "disappointed and disillusioned." He reported that Nixon evidenced an "apparent lack of interest in the space program." Nixon did become emotionally engaged with the fate of the *Apollo 13* crew, but that near-fatal experience only added risk avoidance to lack of interest as part of Nixon's attitude toward lunar missions. For the *Apollo 15* mission in July 1971, Nixon slept through the launch, even though the White House felt it should announce that he had followed the event closely. By that time Nixon was already urging his associates to find ways of canceling the last two Apollo missions, *Apollo 16* and *17*. By April 1970, the iconic "Earthrise" photograph taken during the *Apollo 8* mission that had been hanging on the Oval Office wall throughout 1969 was removed, a symbolic action reflecting the president's lack of commitment to continuing space exploration.

As *Apollo 17* lifted off the lunar surface on December 14, 1972, President Nixon issued a statement saying "this may be the last time in this century that men will walk on the Moon." As the statement was read to the *Apollo 17* crew as they circled to Moon before heading back to Earth, astronaut Harrison "Jack" Schmitt was furious, thinking "that was the stupidest thing a President could have said... Why say that to all the young people in the world... It was just pure loss of will."¹² By his space decisions, Nixon made sure that his forecast would become reality. As of this writing, humans have not traveled beyond the immediate vicinity of Earth for 42 years, and no such journey is planned before 2021, almost 50 years after the last Americans left the Moon.

Nixon coupled his lack of personal interest in continuing Apollo flights to a political judgment with respect to the space program—that the American public was not interested in supporting an expensive, exploration-oriented space program. As he met with NASA Administrator Tom Paine in January 1970 to explain his decision to reject the Space Task Group—recommended post-Apollo program, Nixon told Paine "the polls and the people to whom he talked indicated to him that the mood of the people was for cuts in space."

In May 1961, John Kennedy had paid little attention to poll results showing that the majority of the U.S. public opposed spending the sums of money needed to send Americans to the Moon; Kennedy proposed Apollo as a top-down leadership initiative based on geopolitical considerations. In



Richard Nixon's interest in Apollo missions was not long-lasting. As he met in December 1969 with his assistant Peter Flanigan (at the front of Nixon's desk) and science adviser Lee DuBridge, the famous *Apollo 8* "Earthrise" photograph was hanging on the Oval Office wall. (left image) By April 1970, the photograph was gone. (right image)¹³ (National Archives photo WHPO 2598-15 (left) and WHPO photo 4518-6 (right), the latter courtesy of the Richard Nixon Presidential Library and Museum)

contrast, Richard Nixon saw no persuasive foreign policy or national security reason to lead a reluctant nation and its representatives in Congress toward accepting an ambitious post-Apollo space program, particularly one aimed at developing the capabilities needed for early trips to Mars. Staff assistant Clay Thomas "Tom" Whitehead, who among the White House staff had the most level-headed approach to future space activities, commented that "no compelling reason to push space was ever presented to the White House by NASA or anyone else."¹⁴

The immediate consequence of Richard Nixon's decision not to continue space exploration was suspending production of the Saturn V Moon rocket and approving a NASA budget outlook that forced the agency's leadership to cancel two planned Apollo missions in order to have funds available for future projects. During the 1960s NASA had developed the Saturn V and its related ground facilities on the expectation that the vehicle would remain in service for many years and would be the enabler of a continuing exploration-oriented space effort. These hopes were dashed by Richard Nixon's initial space decisions; those decisions meant that the United States was voluntarily giving up for the foreseeable future the results from its multibillion dollar investment in exploratory capabilities and transforming the unused Saturn V launchers into very impressive museum exhibits.

There is one sense in which Richard Nixon's decision to reject continued space exploration might seem somewhat surprising. Nixon often included space activity as an important aspect of his frequently repeated call for "exploring the unknown," an activity that he believed was essential if the United States was to remain a "great nation." For example, in February 1971 he told a group of astronauts "in the history of great nations, once a nation gives up in the competition to explore the unknown, or once it accepts a position of inferiority, it ceases to be a great nation." In a June

1972 conversation with the *Apollo 16* crew, Nixon equated exploring the unknown with concepts as varying as "science, breakthroughs in education, breakthroughs in technology, breakthroughs in transportation," adding "space—that's the unknown. What's out there?"¹⁵ Nixon did communicate to his associates that he *was* interested in eventual human journeys to Mars, and even mused about the possibility of finding life on a moon of Jupiter, but he saw those activities in the far future, not as objectives related to the decisions he would make during his time in the White House. Nor did Nixon cast his decision to approve the space shuttle in the context of its being an initial step in a decades-long effort to explore destinations beyond the immediate vicinity of Earth. NASA in its input to the Space Task Group had portrayed the space shuttle as part of a coherent long-range strategy ultimately leading to outposts on the Moon and journeys to Mars, and even in 1971 retained elements of that thinking in its technical planning, but that perspective was not considered as part of Nixon's decision to approve the shuttle.

To Nixon, "exploration" was not a sharply defined concept, and his repeated calls for "exploring the unknown" seem to have been little more than what a historian would call a "trope"—an overused rhetorical device offered in the place of substantive thought. Nixon was notoriously poor at conversation with any but those in his inner circle, and falling back on repetitive rhetoric was often his way of dealing with discussions of policy issues with those outside that inner circle. The lack of logic in Nixon's attitude with respect to space activities was on display as he told one of his Congressional relations staff in 1971 that "the United States should not drop out of any competition in a breakthrough in knowledge—exploring the unknown. That's one of the reasons I support the space program." Without pausing, he then said "I don't give a damn about space. I am not one of those space cadets."

Exploring the space frontier was in reality not part of Richard Nixon's strategic vision for America, and thus his repeated call for "exploring the unknown" had little connection to his actual decisions on space policy and budgets in the post-Apollo period. By rejecting the recommendations of the Space Task Group, the Nixon administration attempted to reduce U.S. space ambitions to match the budget it deemed appropriate to allocate to NASA in the post-Apollo period. However, that lowering of ambitions did not happen, either then or since. The exploratory vision still persists; a 2009 blue-ribbon review of the U.S. human space flight program concluded that "Mars is the ultimate destination for human exploration of the inner solar system" and that "human [space] exploration... should advance us as a civilization towards our ultimate goal: charting a path for human expansion into the solar system." Discussing the persistence of this vision, Howard McCurdy suggests "expectations invariably fail, but the underlying vision rarely dies. Rather, people update the vision. The dream moves on."¹⁶

One can argue that Richard Nixon made a major policy mistake in mandating that the space program should be treated as just one of many government programs competing for limited resources. Certainly the belief that this

judgment was ill-conceived is the long-held position of space advocates. But it is also possible that Nixon's decision—that U.S. space ambitions should be adjusted to the funds made available through the normal policy process—was a valid reading of public preferences, and there were no countervailing reasons for him to reject those preferences. If this is the case, then the Nixon administration in its space decisions was correctly reflecting the view of the majority of the U.S. public. There is no evidence that this situation has changed over the past 40 years; the most recent review of the U.S. space exploration program notes "lukewarm public support" for a program of human space exploration and the absence of "a committed, passionate minority large and influential enough" to provide a political basis for such a program.¹⁷

What has actually happened since Richard Nixon made his decisions to end lunar exploration, not to set a new exploratory goal, and to remove the space program's special priority is neither reduced ambitions nor increased budgets; instead, for more than 40 years there has been a continuing mismatch between space ambitions and the resources provided to achieve them. This outcome is close to the worst possible recipe for space program success; *a central part of Richard Nixon's space heritage is thus a U.S. civilian space program continually "straining to do too much with too little."*

Richard Nixon and the Space Shuttle

Although the Nixon decisions to treat the space program as just one of many government activities and to defer human space exploration for the indefinite future have had lasting impacts, the space shuttle program stands as Richard Nixon's most recognized space legacy. Thus any assessment of that legacy must give particular weight to the shuttle's influence on the evolution of the American space program.

Once NASA decided in 1970 to focus on developing the space shuttle as its major post-Apollo effort, there were many designs considered and a number of alternatives to going ahead with a shuttle suggested. During 1971, there was a somewhat confused sorting out of these various possibilities, but as the debate over developing a shuttle reached its final stage, there was little doubt that the White House would approve some version of a shuttle rather than pursue an alternative course. Other options, such as deferring a shuttle decision and carrying out an interim program of human space flight using surplus Apollo hardware or developing an unpowered space glider or a new crew-carrying capsule, had fallen by the wayside. The key decision to be made was thus what kind of shuttle, to carry out what missions, and with what rationale, would be approved.

The options for choice were clearly understood as the decision process reached its climax. As George Low observed in December 1971, "the basic issue on the space shuttle concerns whether or not the shuttle should capture a majority of the payloads that will be flown in the 1980's" and "whether the shuttle should be small or large and whether it should provide for routine operations or one or two flights per year." These two alternative approaches

were embodied in two competing shuttle designs, called here the "NASA shuttle" and the "OMB shuttle." The NASA shuttle—the design ultimately selected—was the end product of more than two years of study by NASA and its aerospace contractors; that study effort had been guided by a combination of national security and NASA's own requirements and the OMB pressure to make the shuttle "cost effective." The NASA shuttle was a full capability vehicle incorporating advanced propulsion, thermal protection, and electronic systems technologies. It would have a 15 × 60 foot payload bay, be able to carry a 65,000 pound payload to a 100 nautical mile orbit due east from the Kennedy Space Center, launch or return a 40,000 pound payload from a polar orbit, and be capable of 1100 nautical miles of cross-range maneuvering. With these capabilities, the NASA shuttle would be able to carry out all planned and potential U.S. civilian, national security, and commercial missions. NASA claimed that it could be launched on a routine basis and at significantly lower cost than any alternative launch system and that it would provide valuable new capabilities for space operations. Such a shuttle, NASA claimed, could be developed for a budget of between \$5 and \$6 billion.

The staffs of the White House Office of Management and Budget (OMB) and Office of Science and Technology (OST) were deeply skeptical of the validity of these claims and indeed of the need for a system with the capabilities NASA was promising. Although many in the two staff offices were indeed skeptical of the value of human space flight itself, they recognized that no American president, and in particular not Richard Nixon, with his emotional view of NASA's astronauts, would choose to end the U.S. human space flight program. They therefore resonated to the advice given by Alexander Flax, chair of the ad hoc panel of the President's Science Advisory Committee set up to examine the NASA shuttle, that "serious consideration must be given to less costly programs which, while they provide considerably less advancement in space capability than the [NASA] shuttle, still continue to maintain options for continuing manned spaceflight activity, enlarge space operational capabilities, and allow for further progress in space technology." This perspective led to OMB recommending to President Nixon in early December 1971 that he direct "OMB and OST to work with NASA on the reorientation of the space program," with a central feature of that reorientation a "smaller, reduced cost" shuttle design. Nixon approved this recommendation on December 3, 1971, and a few days later OMB presented NASA with its concept for a less ambitious shuttle, with a 10 × 30 foot payload bay and 30,000 pound payload lifting capability, to be developed at a budget of no more than \$4 billion.

The question of which of these two alternatives to approve was debated through most of December 1971. Even as the final choice to approve the NASA full capability shuttle was being made over the New Year's weekend, Don Rice, assistant director of OMB, and science adviser Ed David were still arguing strongly against that step. While Rice focused his opposition on the excessive cost of the NASA shuttle, David took a broader view, arguing that

"the large space program implicit in the large shuttle decision is not consistent with the best interests of the nation." The opposition of Rice and David was well-founded and subsequently validated, but they were overruled by their White House bosses and ultimately by President Nixon. Nixon and his associates gave less weight to cost and technical issues than to other political and policy considerations as they decided to approve NASA's preferred shuttle design.

Why Did the Nixon Administration Select the NASA Shuttle?

On November 24, 1971, Richard Nixon had indicated his preference for a "more modest option" with respect to the space shuttle, and again on December 3 had said "yes" to the OMB proposal to work with NASA to examine a smaller, less expensive shuttle design. Yet when the final decision on which shuttle design would be approved was communicated, and most likely made, by George Shultz and Cap Weinberger, it was NASA's full capability shuttle that won the day. The evidence suggests that there were three primary reasons for that choice.

One reason was the importance to the White House of a continuing program of human space flight as a symbol of U.S. space leadership. Cap Weinberger in August 1971 told President Nixon that not moving ahead with some major post-Apollo space effort would be a "mistake," arguing that such a decision "would be confirming, in some respects, a belief that I fear is gaining credence at home and abroad: that our best years are behind us, that we are turning inward, reducing our defense commitments, and voluntarily starting to give up our super-power status, and our desire to maintain our world superiority." Nixon had written on Weinberger's memorandum "I agree with Cap." In making its best case for shuttle approval, NASA had argued "man has learned to fly in space, and man will continue to fly in space. This is fact. And, given this fact, the United States cannot forego its responsibility—to itself and to the free world—to have a part in manned space flight... For the U.S. not to be in space, while others do have men in space, is unthinkable, and a position which America cannot accept." As he met with Fletcher and Low on January 5, 1972, President Nixon asked the NASA leaders if they thought that the space shuttle was a good investment. They of course responded positively, but Nixon, echoing NASA's argument, then added "even if it were not a good investment, we would have to do it anyway, because space flight is here to stay. Men are flying in space now and will continue to fly in space, and we'd best be a part of it."¹⁸ While, as OMB and OST were arguing, a smaller shuttle launched a few times a year would have kept the U.S. human space flight program alive and would have been a useful symbol of U.S. space leadership, clearly a large shuttle launched on a frequent basis would be a much more potent indication of that leadership.

A second reason for the decision to approve the full capability shuttle was President Nixon's interest in its national security uses. John Ehrlichman in

a 1983 interview suggested that "what the military could do with the larger bay in terms of the use of satellites" and the fact that "the space shuttle would have the capability of capturing satellites or recovering them" had "a strong influence on me" and "weighed into my attitude toward the larger shuttle." He added "I feel it is valid to say it also weighed into" the president's evaluation of the shuttle. Nixon's interest in national security uses of the shuttle was well known as final decisions were made, and the president discussed them with Fletcher and Low at their January 5 meeting. As Fletcher and Low met with Cap Weinberger and Jon Rose on March 3, 1972, to discuss NASA's decision on using solid rocket motors to boost the shuttle orbiter off its launch pad, Weinberger reminded the NASA leaders that "the President's strong interest in retaining the military capability" was an important factor in "confirming our decision on the larger size" shuttle.¹⁹

The third and most immediate reason for the choice of the NASA shuttle was the anticipated short-term employment impact of that choice in parallel with Richard Nixon's 1972 reelection bid, particularly in Southern California. The evidence in support of this assertion is conclusive. Aerospace unemployment had emerged as an important political issue in early 1971, particularly after the congressional cancellation of the supersonic transport program. Meeting with Flanigan and Weinberger in May, the president discussed "what could be done about high unemployment areas with specific emphasis on California." Nixon "indicated a very great concern about the California area and the high level of unemployment among technically-trained individuals." The White House created a "California Employment Project" to address this issue in a systematic way. In June 1971, even before the shuttle decision process reached its final stage, Cap Weinberger told a California politician "I am sure that whatever is done will be largely based in California."

On November 24, as he made his initial decision on the space shuttle, President Nixon had commented to Ehrlichman: "Jobs—right, John? Do it in terms of jobs. It ought to be in California." Ehrlichman recalled that the issue of shuttle-related employment in Southern California "was a very important consideration in Nixon's mind... I can recall conversations about that, which were highly persuasive... You must not underemphasize that element, that employment element, in Nixon's decision [on the shuttle], the whole manned space program." When Bill Anders told Ehrlichman in late December that the NASA shuttle would create more short-term jobs than the OMB alternative, the reaction was "OK, that will be the one" chosen. In preparing President Nixon for his January 5 meeting with Fletcher and Low, Flanigan told him that "this program will greatly stimulate the aerospace industry." Nixon himself, discussing his shuttle approval with his political adviser Charles Colson a few days after it was announced, noted that "in Florida and California this [approving the shuttle] is a big deal. It will save the aerospace industry." A few months later, he told a delegation from New York lobbying for the shuttle contract to go to Grumman's Long Island facility that "I'll take the heat for putting the money there [on the space

shuttle contract] rather than the ghettos and all that sort of thing, but then by God we'd better at least get a little credit for it. This is jobs. I mean that is really what is at stake here, jobs."²⁰

Evaluating the Space Shuttle Decision

It was this combination of short and longer-term considerations—the creation of jobs in California before the 1972 election, the interest in potential national security uses of the space shuttle, and the desire to continue a human space flight program that would demonstrate U.S. space leadership to the world and be a source of national pride at home—that led to Richard Nixon's approval of NASA's full capability shuttle. Other factors, such as the ability of the shuttle to operate routinely and at greatly reduced costs, were not greatly influential as Nixon and his top advisers made that choice, even though they became the publicly offered justifications for shuttle development. So an evaluation of the shuttle's impact on the American space program must begin with an assessment of the shuttle program in terms of those objectives that were the proximate reasons for the choice of the NASA-preferred shuttle.

The Shuttle and Human Space Flight

In its 135 flights between April 1981 and July 2011, the space shuttle was undoubtedly the public face of the U.S. space program, communicating to the nation and the world an image of U.S. technological capability and American leadership. The shuttle orbiters carried 355 different people into orbit, including 306 men and 49 women, with many making multiple flights; two U.S. astronauts each flew on seven shuttle missions. The relatively nonstressful conditions of launching aboard the shuttle opened up the experience of space flight to scientists and engineers, and also to a few politicians, teachers, and industry representatives, not just to test pilots. Astronauts from 16 countries flew aboard the shuttle, thereby fulfilling Richard Nixon's "pet idea" of flying non-U.S. people on a U.S. spacecraft. (In fact, while Nixon wanted only the symbolism connected with flying non-Americans on a NASA spacecraft, his interest opened the door to intimate international participation in the U.S. human space flight program, leading to the European Spacelab effort and the Canadian robotic arm on the shuttle, the U.S.-Soviet *Apollo-Soyuz* Test Project, and eventually to the 15-partner International Space Station (ISS).) Of the shuttle's 135 missions, 37 were dedicated to assembling and outfitting the ISS; maintaining the capability to launch space station elements had been a NASA "bottom line" in the final stages of the shuttle debate. Demonstrating the unique capabilities offered by the shuttle, other missions launched, repaired, and recovered satellites, most notably the Hubble Space Telescope, sent probes to the Sun, Venus, and Jupiter, launched other telescopes to observe the universe, and hosted on-orbit research. There were nine shuttle dockings with the Soviet/Russian

space station *Mir*. Unfortunately, two missions ended in catastrophe; in each, seven crew members lost their lives.

The shuttle was and continues to be a source of considerable pride for U.S. citizens. Images of a shuttle launch are global symbols of American accomplishment and technological leadership, and even after they have been retired from service the three remaining shuttle orbiters—*Discovery*, *Atlantis*, and *Endeavour*—are objects with high public appeal. In terms of its political impact and of offering unmatched capabilities for space operations, the space shuttle was a resounding success. The space shuttle met the objective of keeping Americans flying in space as a source of national prestige and pride; the capabilities offered by the shuttle made the United States the unquestioned leader in human space flight.

National Security Uses of the Space Shuttle

Another of the influences on the choice of the full capability shuttle was President Nixon's interest in its ability to launch the most advanced intelligence satellites and to carry out innovative national security missions. Those missions included the shuttle launching on demand during a political or military crisis, conducting a single-orbit satellite deployment or rendezvous, or inspecting or even destroying a potentially hostile satellite.

While the president himself may have been attracted by such national security uses, the reality was that support for the shuttle within the military and intelligence community was at best tepid, both at the time the shuttle decision was made and afterwards. Deputy Secretary of Defense David Packard's 1971 flexibility on shuttle requirements is suggestive of an ambivalent Department of Defense attitude toward the vehicle, and the effort in late 1971 to get a joint NASA-DOD statement to the president in support of the shuttle apparently did not bear fruit. During 1972 Congressional hearings on the shuttle program, DOD and Air Force testimony was supportive but guarded in character; the military took the position that the DOD would commit to depending on the shuttle only after its capabilities and constant availability had been fully demonstrated. During the mid-1970s top-level Department of Defense support for the shuttle ebbed and flowed. At lower levels of the national security community, there was strong opposition to phasing out expendable launch vehicles until the shuttle was demonstrated to be completely reliable. The DOD did agree to pay the costs of a west coast launch site for the shuttle at Vandenberg Air Force Base, since that location was primarily needed for national security launches into polar orbit. In addition, DOD agreed to be responsible for funding the "space tug" to move payloads from the shuttle to higher orbits and for covering the costs of separate launch control centers in Houston and Colorado Springs for managing classified shuttle missions. With the urging of Hans Mark, first as Undersecretary and then as Secretary of the Air Force from 1977 to 1981, and for much of that time also director of the National Reconnaissance Office (NRO), some national security satellites were redesigned to take advantage of the shuttle's

attributes. When in 1979 President Jimmy Carter considered canceling the shuttle program because of its cost overruns, it was the national security uses of the shuttle, particularly in terms of launching the photo-reconnaissance satellites needed to verify arms control agreements, that convinced the president to continue the program. Once the Reagan administration took office in 1981, an early action was to confirm as national policy that the shuttle would be "the primary space launch system for both United States military and civilian government missions."²¹

This policy declaration represented the high point of the notion of using the shuttle for national security missions. Within the first years of the Reagan administration, Air Force and NRO resistance to total dependence on the shuttle escalated into a conflict that required a presidential decision to resolve. The consequences of total U.S. dependence on the shuttle had been predicted. In the midst of the shuttle debate in 1971, the OMB had warned "for national security purposes, we may not want all our eggs in one basket." The Air Force and NRO in 1984 won the right to develop an expendable launch vehicle as a backup to the shuttle for the largest national security payloads; this turned out to be the Titan IV booster. In the aftermath of the January 1986 *Challenger* accident, most national security payloads were removed from the shuttle and expendable launch vehicle production lines were activated; the nearly complete multibillion dollar West Coast launch site for the shuttle was mothballed.

Only ten dedicated national security missions, eight of which were classified, were launched aboard the space shuttle, including eight missions after the 1986 *Challenger* accident; the payloads for most of those missions had been uniquely designed for shuttle launch. Some of the capabilities relevant to national security uses, such as satellite repair, recovery, and refueling, were demonstrated on other early shuttle missions. But as a national security system, the shuttle had no continuing utility. One historian of national security space activities cites a Department of Defense 1992 report that set the cost of redesigning military and reconnaissance spacecraft first to launch on the shuttle and then reconfiguring them again to launch on the expensive Titan IV expendable launch vehicles after the *Challenger* accident as "in excess of \$20 billion."²² None of the ten national security shuttle missions required the cross-range capability that had been an original DOD demand, and none of the innovative missions described in the 1969 DOD/NASA space shuttle report that had influenced Richard Nixon's support of the NASA shuttle were ever attempted. Rather than provide new capabilities used by the national security community, the shuttle turned out to be a multibillion dollar drain on the national security space budget.

The Space Shuttle and Aerospace Employment

The space shuttle prime contract was awarded in mid-1972 to North American Rockwell, a company with its space operations based in Southern California. This award meant that the projected California employment impacts, both

in advance of the 1972 presidential election and subsequently, were achieved. Although Rockwell barely beat out New York-based Grumman Aerospace for the contract award, there has been no evidence discovered in the course of research for this study that Richard Nixon's expressed wish to put a large share of shuttle work in California and his personal relationship with Willard "Al" Rockwell, the head of North American Rockwell, translated into an overt White House attempt to influence NASA as it selected the shuttle prime contractor. But NASA certainly was fully aware of the president's interest as that decision was made.

Basing shuttle approval on its job-creating impact set an unfortunate precedent for many subsequent space decisions. (In 1961, the politically driven decision to locate the Manned Spacecraft Center in Houston, Texas, as a new NASA facility for the Apollo program was a forerunner of this precedent.) From 1972 on, the employment and institutional impacts of various space program choices have been an important, sometimes overriding, factor in reaching a decision on how to proceed. This is especially the case since most decisions on large space projects since 1970 have been made through the normal political process, where such parochial considerations play a significant role. The widely accepted view of the civilian space effort as a "jobs program" had its origins in the Nixon administration's decision to "save the aerospace industry" by approving development of the full capability shuttle. The job-maintenance or job-creation impact of various space program options continues as a strong influence on twenty-first century decisions.

In terms of the proximate reasons for its White House approval, then, the space shuttle program must thus be judged a mixture of success and disappointment. In particular, the shuttle during its three decades of operation served the nation well as a focus for U.S. space leadership and the resultant prestige and pride. In terms of its role in U.S. military and intelligence efforts in space, some of the still classified national security missions launched aboard the shuttle are likely to have produced useful results, but overall the space shuttle program turned out to be a very expensive detour for the national security space program. The shuttle program's success in producing aerospace jobs in advance of the 1972 election and in the longer term helping revitalize the aerospace industry has been a mixed blessing; it achieved Richard Nixon's short-term political objectives while creating the image of the space program as a "public works" effort.

What about the Other Reasons for Shuttle Approval?

There were other reasons put forward for going ahead with the full capability shuttle, and the success or failure of the shuttle in satisfying those reasons must be included in an overall assessment of the shuttle as a Nixon space legacy. Although secondary factors in the presidential-level decision to approve NASA's full capability shuttle, the claims that NASA had made from 1969 on—that the shuttle could be launched on a routine basis and that it would significantly lower the costs of space operations—became the

publicly offered reasons for developing the shuttle. The space shuttle failed to match those claims. As then-NASA Administrator Mike Griffin commented in 2010, "what the shuttle does is stunning, but it is stunningly less than what was predicted."²³

In his prepared statement announcing shuttle approval, President Nixon said that the shuttle would "revolutionize transportation into near space by routinizing it." At the time the shuttle decision was made, the intent was to launch the shuttle 40 to 60 times per year once it became fully operational. This of course never happened. The most launches of the shuttle in a single calendar year ended up being nine in 1985, with the average annual launch rate over the 30-year lifetime of the program being 4.3 launches per year. The shuttle could not be launched on demand; rather, it took a lengthy and labor-intensive process to prepare each shuttle mission for launch. Rather than a vehicle capable of frequent and routine operation, the shuttle turned out to be, in the words of the Columbia Accident Investigation Board, a "complex and risky system" and a first generation "developmental vehicle" which required great care to operate safely.²⁴

The Nixon January statement also said that the space shuttle would "take the astronomical costs out of astronautics." In March 1972, as the final shuttle design was announced, NASA Administrator Jim Fletcher claimed that once the shuttle became operational, its incremental cost per launch would be \$10.5 million (almost \$60 million in 2014 dollars). But NASA in 2012 estimated the cost of each shuttle launch, depending on how it was calculated, as between \$814 million and \$1.266 billion, up to 20 times higher than the 1972 estimate. NASA in December 1971 had said that the full-capability shuttle would carry payload into orbit at a cost of \$118 per pound (\$691 in 2014 dollars); in Congressional testimony, NASA listed the cost of using the shuttle to deliver cargo to the space station as \$21,268 per pound in 2011 dollars.²⁵ By any measure, the shuttle did not "take the astronomical costs out of astronautics."

One may ask why NASA's 1972 estimates of shuttle operating costs were so far off the mark, while the estimates of shuttle development costs made at the same time were close to the actual amount. NASA cost estimators had significant experience in forecasting the costs to develop systems such as the Mercury, Gemini, and Apollo spacecraft, various launch vehicles, and scientific spacecraft. Based on that experience, they were able to estimate the cost of designing, developing, and testing the final NASA shuttle design with enough precision to allow Fletcher and Low to promise the White House that the cost was likely to be approximately \$5 billion and certainly not more than \$6 billion. The actual development cost was \$5.5 billion in 1971 dollars, which were the basis for the NASA cost estimate.

None of those involved in the shuttle decision were as concerned about operating costs as they were about keeping the annual budget and total cost of shuttle development below politically acceptable levels. There was little White House or NASA leadership attention given to the quality of the cost-per-launch forecasts being put forward. NASA had little experience in

estimating the costs of repetitive operation of a space system, and thus its estimates of shuttle operating costs were very uncertain. NASA, coming off its Apollo successes, was a technologically confident, perhaps even arrogant, organization, believing that it could incorporate advanced technology in such areas as propulsion, thermal protection, and on-board electronic systems into the shuttle design in ways that would make the vehicle able to be operated inexpensively and on a routine basis. The per flight cost estimate provided to the NASA leadership in late 1971 was based on assumptions of 50 flights per year, vehicle self-checkout, fully reusable thermal protection, and long engine life without major repair. While that estimate may have represented the best NASA could do in 1971, it ended up being based on the wrong technological assumptions and far off the mark. The shuttle required many hours of human labor for its checkout, a number of the shuttle's thermal protection tiles had to be repaired or replaced after each mission, and the shuttle's main engines required extensive refurbishment between uses. Shuttle program manager Robert Thompson characterized the NASA cost per flight estimate as an "optimistic guess." That "guess" was nowhere near the actual cost once the shuttle began flying; rather than the shuttle significantly lowering the cost of space launch, it became an extremely expensive system to operate. Because of the need for extensive checkout and refurbishment between flights, the shuttle that was developed had no chance of being operated at a 40 or more launches per year flight rate; in addition, the anticipated demand for that many shuttle launches a year never materialized.²⁶

The Space Shuttle and the Space Station

From 1970 on, one of the performance requirements driving space shuttle design was NASA's intent at some future point to use the shuttle to launch elements of a space station. This was recognized by the OMB staff, who observed in 1971 that "in a sense, a commitment to a shuttle is an implicit commitment to a subsequent space station program." There is no evidence that this shuttle-station link was considered by the president and his senior advisers as the final shuttle decision was made, but the choice of the NASA shuttle design carried with it the virtual certainty that a future president would be asked to approve a shuttle-launched station.

That is precisely what happened. The shuttle's first flight was in April 1981; soon after that flight, President Ronald Reagan's nominees for NASA administrator and deputy administrator, James Beggs and Hans Mark, agreed that they "would try to persuade the new administration to adopt the construction of a permanently manned space station as the next major goal in space." The two announced their intent at their Congressional confirmation hearing in June 1981, in essence repeating Tom Paine's 1969 argument that the space station was "the next major evolutionary step in man's experimentation, conquest, and use of space." Beggs and Mark characterized the station as "the next logical step." It took almost three years for NASA to gain presidential approval; during his State of the Union address on January

25, 1984, Ronald Reagan announced that "I am directing NASA to develop a permanently manned space station and to do it within a decade."²⁷

Discussing the long and troubled history of the space station project is beyond the scope of this study; the point here is that from its 1968 origin as the logistics vehicle for a Saturn V-launched space station, through the 1970 decision to switch to a shuttle-launched station and then to defer station development until the shuttle was flying, to the final July 2011 outfitting mission to what had become the International Space Station, there was an unbreakable link between the shuttle and the station. That bond meant that, unless the station program was terminated early, NASA had to keep the shuttle in service until station assembly and outfitting were completed. The high costs of the shuttle and station programs thus dominated the NASA human space flight budget for almost 40 years.

Was the Space Shuttle a "Policy Failure"?

In an article published soon after the 1986 *Challenger* accident, I suggested that "the space shuttle program must be assessed as a policy failure, at least in terms of meeting the objectives [lower cost and routine operation] that have been its articulated rationale since 1972." In deciding to approve the NASA shuttle, "too much attention was paid to the short term, while longer range considerations were inadequately considered... The shuttle decision stands as a powerful example of how not to make a national commitment to an undertaking on which many other significant projects depend."²⁸ Do these judgments still stand up, almost three decades later? Was the shuttle *program* itself a failure? Or was it the Nixon administration *decision* to approve the NASA full capability space shuttle that was the policy failure? I was not very clear in what I wrote in 1986, but it was my judgment then, and now, that the latter alternative is the case. As the preceding paragraphs have suggested, the record of the space shuttle program is a mixture of success and failure. But there were in 1971 better alternatives to approving development of the NASA full capability shuttle, and thus that approval is better described as a policy mistake, rather than a policy failure.

The Roots of the Policy Mistake

The policy mistake in the decision to develop the full capability space shuttle had deep roots in the history of the space shuttle program. The 2003 report of the Columbia Accident Investigation Board concluded that "the greatest compromise NASA made was not so much with any particular element of the technical design, but rather with the premise of the vehicle itself. NASA promised it could develop a Shuttle that would be launched almost on demand and would fly many missions each year." The report added "the increased complexity of a Shuttle designed to be all things to all people created inherently greater risks than if more realistic technical goals had been set from the start. Designing a reusable spacecraft that is cost-effective is a

daunting engineering challenge; doing so on a tightly constrained budget is even more difficult.²⁹ That was the situation in which NASA found itself in 1970 and 1971, but NASA's leaders persisted in their advocacy of the full capability shuttle, even as some of them, particularly George Low, questioned the wisdom of that advocacy.

There were actually two policy mistakes associated with the shuttle decision. *The first and more fundamental mistake was the White House accepting as the basis for its shuttle decision NASA's claim that it could successfully go directly from the Apollo program, characterized by brute force launcher technology and crew-carrying capsules parachuting to an ocean landing, to developing a highly capable vehicle in terms of payload capacity, in-orbit operation, and maneuvering during entry, incorporating advanced technology in many of its systems, with a high degree of reusability, and able to land on a runway and quickly be readied for another launch, all at relatively modest cost compared to the alternatives.* The bullish vision of people such as Tom Paine and George Mueller pushed NASA to focus on an ambitious shuttle design incorporating advanced technology and capable of "air-line type" operations. There was a significant degree of technological hubris in NASA's view of what would be achievable. After all, NASA engineers and managers had just succeeded in landing American astronauts on the Moon and were convinced that they could overcome the next set of technological challenges.

NASA and its contractors thus focused their attention during 1970 and the first half of 1971 on finding the best design meeting NASA and national security community requirements and employing cutting edge technology in areas such as propulsion, thermal protection, and onboard electronic systems. After May 1971 they had to carry out their design studies within an OMB-imposed budget ceiling in terms of both peak annual funding and the overall cost of the shuttle program. Although NASA recognized by mid-1971 that a two-stage fully reusable shuttle design was not feasible either financially or technologically, there was little emphasis on investigating less ambitious, less expensive, alternatives to an advanced technology shuttle orbiter with a variety of means for boosting it into space. There was essentially no attention given at the engineering level to concepts such as the glider favored by the Flax committee or the smaller shuttle proposed by OMB, or even to the Mark I, less technologically ambitious, shuttle proposed by NASA Headquarters.

In addition to designing a shuttle that could be developed within a constrained budget, NASA engineers were forced into demonstrating the shuttle's overall cost-effectiveness. In 1970, the Bureau of the Budget and then its successor OMB had insisted on proof that the shuttle development and operation would cost less than using expendable vehicles to launch U.S. space missions. NASA concluded that it had to satisfy that unprecedented OMB requirement. Demonstrating the shuttle's cost-effectiveness became perceived as a political necessity, and likely led to NASA's leaders and engineers deluding themselves about the costs of operating the

shuttle on a frequent basis in order to make the economic case come out positively.

The design ultimately recommended was likely the best engineering solution to the demanding requirements NASA's technical staff was asked to meet. But that design created a first-generation experimental vehicle, not a shuttle capable of delivering the cost savings and routine operational benefits that NASA was promising. Basing the White House decision to approve the NASA shuttle on other factors while implicitly accepting NASA's optimistic claims with respect to the shuttle's operation was a policy mistake with long-lasting consequences.

Were There Alternative Choices Available?

There were some individuals both inside and outside of NASA who recognized the difficulties in pursuing the full capability shuttle as NASA's immediate post-Apollo project. NASA's top spacecraft designer, Max Faget, argued as the shuttle program was gaining momentum in early 1969 that a first step should be a relatively small vehicle, which he characterized as the space equivalent of the first-generation DC-3 commercial aircraft. Secretary of the Air Force Robert Seamans during the Space Task Group deliberations suggested that "it is not yet clear that we have the technology" for a reusable space transportation system that would produce major reductions in the cost of transporting payloads into space, and suggested "a program to study by experimental means including orbital tests" the feasibility of such a system. As NASA awarded shuttle design study contracts in 1970, veteran flight director and then-deputy director of NASA's Manned Spacecraft Center Chris Kraft warned "I don't think we should try and build an ultimate vehicle the first time... I think we've got to be extremely careful that we don't try to build a do-all vehicle. I don't think we ought to talk ourselves into the fact that the shuttle is to do every job in the space program."³⁰ The idea that there should be an interim step before deciding whether to develop a full-capability shuttle was present throughout the shuttle debate, but was never embraced by those directly in charge of the shuttle program, who remained convinced that they could design and develop the kind of shuttle approximating the advanced technology vehicle first suggested by George Mueller.

NASA's leaders themselves, as the final decision on the space shuttle approached, harbored reservations about the viability of the full capability concept. George Low in August 1971, as he assessed alternative courses of action, concluded that "we should drop the shuttle right now and come up with a different manned space flight program." He added that "this program should be based on an evolutionary space station development, leading from Skylab through a series of research and applications modules to a distant goal of a permanent space station. We should also set for ourselves a distant goal of a lunar base. The transportation system for this manned space flight program would consist of Apollo hardware for Skylab; a glider launched on an expendable booster for the research and application modules; and finally, the

shuttle but delayed 5 to 10 years beyond our present thinking." As Low and Jim Fletcher prepared the NASA budget request due at OMB on September 30, 1971, they "debated whether we should not forego the shuttle entirely and develop instead some alternate manned space flight program."

Fletcher and Low did not choose this option, getting little support for it from the NASA technical workforce and deciding that it was in NASA's institutional interests to seek immediate approval of an ambitious shuttle design. This was a fateful decision, since it polarized the shuttle debate; during the rest of 1971 NASA would make its case for approving the full-capability shuttle as a "best buy" rather than seek a compromise with shuttle skeptics in OMB and OST. The decision-making process functioned as it should, elevating two shuttle options for presidential choice. *But as the White House made the final shuttle decision, Nixon and his top advisers chose the wrong option. This was the second policy mistake connected to the space shuttle decision.*

That going ahead with the full-capability shuttle was a course of action fraught with the potential for future problems was clear to some of those examining shuttle choices. For example, Alexander Flax had reported to science adviser David in October that "most of the members of the Panel doubt that a viable program can be undertaken without a degree of national commitment over the long term analogous to that which sustained the Apollo program. Such a degree of political and public support may be attainable, but it is certainly not now apparent." Flax added "planning a program as large and as risky (with respect to both technology and cost) as the shuttle, with a long-term prospect of fixed ceiling budgets for the program and NASA as a whole does not bode well for the future." This was prescient advice, but it was not heeded.

The commitment to NASA's full-capability shuttle (which carried with it a future decision to develop a space station) created for more than four decades two very expensive "mortgages" on the NASA annual budget. Given that that budget was commanding a decreasing share of federal discretionary spending, the necessity of servicing these mortgages meant that there were limited funds available for other worthy space endeavors. As Bill Anders, a veteran of the shuttle decision process, recently commented, "the shuttle, like a cuckoo in the nest, pushed out many less sexy but higher pay-off science and commercial programs for lack of funds."³¹

It is of course impossible to know what might have happened if the White House had chosen the OMB shuttle option. But it does seem that pursuing a less ambitious shuttle design as an intermediate step in the evolution of U.S. space capability might well have made more technical sense and could have initiated an evolutionary U.S. space program that would have been a better fit to the resources that the political system has made available to NASA over the past four decades.

There is another piece of evidence suggesting that the 1972 decision to develop the full-capability orbiter was a policy mistake. The absence in the wake of the shuttle's 2011 retirement of any advocacy within the U.S. space community for replacing the NASA shuttle with a second generation system

having similar or greater capabilities is striking. After 30 years of operating the shuttle, there is no current demand to replicate in one vehicle the capabilities that the shuttle provided.³²

My assessment of the space shuttle *program* as a major element of the Nixon space legacy is a mixture of positives and negatives. It is a matter of judgment whether the former outweigh the latter. But I stand by my 1986 assessment that the *decision* to develop the full capability shuttle was indeed a "policy failure," better characterized as a "policy mistake," in that the consequences of that choice have had a strongly negative impact on the evolution of the U.S. space program. Jumping directly from Apollo to developing the full capability shuttle was "a leap too far." Rather than being a cost-effective system providing highly valued capabilities, the shuttle turned out to be an expensive and difficult to operate vehicle. Arguing that the shuttle enabled the United States to develop the International Space Station is somewhat circular, since it is not yet clear that the station will turn out to produce benefits worth its development cost, and most likely there would not have been a space station (or at least the station that was constructed) without the shuttle.

The Nixon Space Heritage

The space shuttle may be the most visible Nixon space legacy, but the consequences of the other Nixon administration decisions in the 1969–1972 period have also had pervasive and lasting impacts. A 2012 assessment of NASA's "strategic direction" observed that:

The National Aeronautics and Space Administration (NASA) is at a transitional point in its history... The agency's budget... is under considerable stress, servicing increasingly expensive missions and a large, aging infrastructure established at the height of the Apollo program. Other than the long-range goal of sending humans to Mars, there is no strong, compelling national vision for the human spaceflight program, which is arguably the centerpiece of NASA's spectrum of mission areas. The lack of national consensus on NASA's most publicly visible mission, along with out-year budget uncertainty, has resulted in the lack of strategic focus necessary for national agencies operating in today's budgetary reality. As a result, NASA's distribution of resources may be out of sync with what it can achieve relative to what it has been asked to do.

This review concluded that "there is no national consensus on strategic goals and objectives for NASA."³³ This judgment was echoed in the most recent review of the human space flight program, which observed that "a national consensus on the long-term future of human spaceflight... remains elusive."³⁴

To a significant degree this unsatisfactory condition of the U.S. human space flight program in the second decade of the twenty-first century is a heritage of the policy decisions made by Richard Nixon more than 40 years

ago. Approving the space shuttle came without a meaningful national commitment to post-Apollo space program objectives—there was no “strategic focus.” George Low in October 1970 had suggested that “with the shuttle the U.S. can have a continuing program of manned space flight... without a commitment to a major new manned mission goal.” This proved to be a winning argument; by approving the space shuttle, a capability-justified means for carrying out a variety of space activities, Richard Nixon avoided having to define the long-term space objectives which the shuttle would serve. This lack of guiding goals for the U.S. space program has persisted for more than 40 years, causing many to characterize the program as “adrift.” If this characterization is accepted, it was Richard Nixon that set NASA on that goal-less voyage.

Nixon and his closest advisers gave little attention to the longer term consequences of their decision to put the NASA full-capability space shuttle at the center of the post-Apollo space program. Those consequences were compounded by approving a shuttle design that from NASA’s standpoint was a step toward an eventual space station. The consequences were exacerbated by setting out an approach to determining the NASA budget that was very likely to result in funding insufficient to support efficient development and operation of both the space shuttle and space station while also funding the space activities they were designed to serve. It was difficult to rally public and political support for the capability-driven approach inherent in the Nixon approach to the post-Apollo space program, and the lack of broad public support for the space program has persisted. The absence of a compelling exploration objective or other widely accepted goal has resulted for four decades in a human spaceflight program focused, for uncertain purposes, on developing and operating the shuttle and assembling the space station. Attempts in 1989 and 2004 to set an exploration goal to guide the space program have not taken root, and the fate of the current NASA exploration program is unclear.

There is no simple or immediate remedy to the current situation with respect to the U.S. space program. It will be very difficult to undo the consequences of flawed or mistaken policy decisions made more than four decades ago. Some suggest that the government should step aside and allow the U.S. private sector to take the leading role in the U.S. space program, including human exploration. My judgment is that such an approach is unrealistic; only governments can provide resources sufficient to lead the initial stages of a long-term exploration effort, although government-private sector partnerships (and international cooperation) should certainly be part of that effort. In my view, if the United States is to remain the leader in human space exploration it will take committed and continuing presidential leadership of the character provided so long ago by John F. Kennedy, once again singling out the space program for special priority and setting challenging goals, convincing a reluctant public and their representatives in Congress to accept those goals, and then, crucially, committing on a sustained basis the political, human, and financial resources needed to achieve them.³⁵ The alternatives

are to continue to drift along, trying to do too much with too little, or, less likely, to lower U.S. ambitions in space to match the funding available. A comprehensive review of the U.S. space program in 2014 once again concluded, as had its 2009 and 2012 predecessors, that “the human spaceflight program conducted by the U.S. government today has no strong direction” and that “the long-term future of human spaceflight... is unclear.”³⁶

That situation is Richard Nixon’s most fundamental space heritage.